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Conscious Computers or Completely Dark

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Origins and Meaning

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Feel the air enter your lungs, the screen you are looking at, and the noises that enter your ear. Beyond the physical characteristics of the air, the screen, the noise, and their composition or temperature, they all *feel* like something. Your mind takes their objective information and produces a subjective feel. The vividness of the screen, the fullness of your breath, and the serenity of the halls all produce an introspective feeling. In essence, there is a certain *feel* to being you: philosophers call this qualia or phenomenal consciousness. However, do other people and things have this subjective feeling? Do pebbles, zucchinis, your best friend, or dogs have subjective consciousness? Most would probably say no to the first three and yes to the last two but why? Is it because the first two are too different to ourselves? But what are the relevant attributes something must possess for us to grant them phenomenal consciousness? Is intelligence an important attribute? This feels right: dogs are smarter than rocks but dumber than humans. While this “intelligence” account seems reasonable, it begs the question of how intelligence is defined.

In the face of such rapid technological advancement in artificial intelligence this paper will uncover the inner workings of current AI’s and explore if they are conscious. With an examination of Turing’s Imitation Game, Searle’s Chinese Room argument, and Chalmer’s phenomenal zombies, it will demonstrate how not only are current LLM not conscious or intelligent, but that intelligence is insufficient for consciousness. Then, this paper will look back in time and weigh the advantages and disadvantages of various theories of consciousness from Cartesian Dualism to Integrated Information Theory. Finally, this paper will explain how to proceed amidst the current epistemological uncertainty, and the moral implications of conscious AI’s.

Recent large language models (LLM) such as OpenAI’s ChatGPT and Google’s Gemini, satisfy nearly all outward lay definitions of intelligence. They can solve competition level math or coding problems better than all but the best in this field, persuade people on all

topics better than the average human, and provide valuable information in a conversation on virtually all topics... These LLM's have even passed the famous "Imitation Game", which the "Father of Computation" British philosopher and computer scientist Alan Turing devised (Scott). According to Turing, a computer or machine is said to have achieved human intelligence and can think if it passes the "imitation game". For this game, it requires three participants: a human player, a human interrogator, and a machine player. All three participants are in separate rooms and the interrogator, who does not know who the human is, poses questions to either player and tries to determine which player is the machine from their answers. Some possible topics of questions are chess, hair length, and arithmetic. The more a machine can pass as a human, the more intelligent and capable of thought it is said to be. (Turing 433-438)

The imitation game reflects Turing's underlying philosophical view on the philosophy of mind. As the earliest proponent of the modern functional Computational theory of mind (CTM), Turing posits that minds are a form of informational processing system.¹ By informational processing system, this theory suggests that brains are a type of machine that processes sensory and internal information in accordance with some "brain" software or algorithm(Gennaro), which is "A well-ordered collection of unambiguous and effectively computable operations that when executed produces a result and halts in a finite amount of time" (Blair). Like the ones and zeros that computers work with, the sensory and internal information do not have intrinsic meaning to us but rather appear to have meaning when our brain follows a program and manipulates or processes them.

Searle's Chinese room thought experiment exposes the shortcomings of functionalism and the CTM. It does so by illuminating how the passage of the imitation game is inadequate

¹ Functionalism in the context of philosophy of mind is the position that mental states or minds are wholly constituted by their function or role in a larger system rather than their material makeup.

for a machine to be bestowed with intelligence in the way that Turing desired. The premise is as follows:

Imagine a native English speaker who knows no Chinese locked in a room full of boxes of Chinese symbols (a database) together with a book of instructions for manipulating the symbols (the program). Imagine that people outside the room send in other Chinese symbols which, unknown to the person in the room, are questions in Chinese (the input). And imagine that by following the instructions in the program the man in the room is able to pass out Chinese symbols which are correct answers to the questions (the output). The program enables the person in the room to pass the Turing Test for understanding Chinese but he does not understand a word of Chinese. (Searle 115-116)

For Searle, the thought experiment illustrates that our mind and intelligence are more than just the manipulation of inputs and outputs: true intelligence is invariably tied with understanding. By training on billions of examples of human-written text, the model learns statistical patterns about how language is structured. Rather than memorizing sentences or counting word pairs, it learns to represent words and phrases as vectors in a high-dimensional space, where meanings that behave similarly across contexts end up close together. When you ask for a “recipe for banana bread,” the model does not retrieve past recipes; instead, it transforms your prompt through many layers that have learned how instructions, ingredients, and cooking language are typically organized, and then generates text that follows those learned patterns. Thus, while humans may not be manually writing the code behind ChatGPT or even understand every line, the machine is only manipulating data (words) in accordance with a set of rules, and nothing more. The system is Searle’s Chinese room: seemingly

externally intelligent but internally dark as no mental or semantic content accompanies each output.

Thus, while systems like ChatGPT are often described as “intelligent,” their operation differs fundamentally from human understanding. Humans do not manually encode the system’s responses, nor do they specify explicit rules for how language should be used. Instead, the model is trained on vast amounts of human-written text using self-supervised learning, during which fixed algorithms adjust numerical parameters to capture statistical regularities in language. At inference time, the system does not retrieve examples or reason about meanings in a conscious sense. It transforms tokenized inputs into high-dimensional numerical representations and repeatedly applies learned mathematical operations to produce a probability distribution over possible next tokens. These operations can reliably reproduce patterns of syntax, argument structure, and discourse that humans recognize as meaningful. However, this competence arises from learned transformations of representations, not from awareness or experience. This distinction is crucial when comparing such systems to John Searle’s Chinese Room. Like the agent in the room, the model produces outputs that are indistinguishable from those of a fluent speaker, despite lacking any subjective grasp of what its symbols represent. There is no accompanying experience of meaning, intention, or understanding. The system does not know what a “recipe,” a “banana,” or “bread” is; it has no sensory grounding, goals, or point of view.

Indeed, potentially the most famous living philosophy of mind philosopher, David Chalmers, agrees with Searle that our conscious minds are more than just their functions. For one, there are “easy problems of consciousness”, which are susceptible to normal neuroscience inquiries (Chalmers 200). These include “the ability to discriminate, categorize, and react to environmental stimuli”, “the integration of information by a cognitive system”, “the ability of a system to access its own internal states”...(Chalmers 200-203). The hard

problem of consciousness, however, is explaining qualia. According to Chalmers, the reason why the hard problem is hard is that it “goes beyond problems of the performance of functions” (Chalmers 204). Even if we gain a complete understanding of the physical processes of our brain, all of the easy problems like “perceptual discrimination, categorization, internal access, verbal report”, the hard problem may still be unanswered: “Why is the performance of these functions accompanied by experience”(Chalmers 204-205). This is because all easy problems are inherently physical while qualia is non-physical and subjective, thus unable to be fully captured by functional analyses. To further illustrate why solving easy problems is not necessarily sufficient for solving the hard problem, let’s consider zombies.

Phenomenal zombies are creatures that are identical to us in every functional way, but lack consciousness or qualia. When they taste a fresh apple or see an orange sunset, they respond in the same way as us: remarking on the apple’s juiciness or the vividness of the hues. However, they are internally dark and lack any subjective feel of life, of consciousness. Now, if functionalism was true and qualia could simply be explained by explaining the different functions and processes of one’s mind, such a creature would be inconceivable. However, while such a creature may not really exist, it is conceivable. The case for the conceivability of zombies is evident by the popularity of the “other minds” problem² and spectrum inversion.³ Therefore, the hard problem remains hard—in the sense that it cannot be resolved by resolving easy problems—so long as we view zombies as a conceivable non-contradictory idea. (Kirk)

Is intelligence completely beside the question when discussing consciousness? No.

While Searle’s discussion was superficially about intelligence, it veers into our common

² The Problem of Other minds posits that since we can never directly experience or read the minds of others, but must instead observe their behavior, how can one know for certain that others have minds and thus can think. In video game terms, how do you know that you are not the only player and that everyone else is just NPC’s.

³ The Spectrum inversion thought experiment simply suggests that it is possible for my internal reception of the color red to be your blue.

understanding of consciousness to a large degree. We implicitly define understanding anthropocentrically with the inclusion of semantic meaning: the manipulation of symbols following a set of rules is not understanding because there seems to be more that's going on, but what is this externality? The answer is qualia or phenomenal consciousness. After all, it isn't entirely clear to me how semantic meaning differs from subjective experience, albeit limited to words. Thus, if we say that consciousness arises from intelligence but intelligence requires semantic meaning, which itself requires consciousness, we are begging the question. Consciousness does not depend on intelligence but intelligence depends on consciousness. For even a child, or a mentally handicapped adult is conscious, which is to say that there is something it feels like to be them. Therefore, intelligence is the product of consciousness and to truly uncover the causes of qualia, we must not fixate on intelligence. Having established the importance of consciousness in relation to machine intelligence, let us tackle the hard problem.

In Rene Descartes's *Meditations*, he introduces the metaphysical view of dualism. In this view, the world is composed of two types of things: mind and body. The mind or the soul is non-spatial, indivisible, conscious and thinking, and subjective while the body is spatial, divisible, non-conscious, and objective. (Descartes 64-68) Humans are just minds connected with bodies. Therefore, there isn't really any hard problem of consciousness per se because dualists do not attempt to prove that consciousness arises from any physical process. Rather, phenomenal consciousness arises from the possession of the mind, and zombies simply would not exist. While this "solves" the hard problem, it has many drawbacks, chief of which is the interaction problem.

In a letter to Descartes, Princess Elisabeth of Bohemia raises the interaction problem. If minds are non-spatial and therefore lack physical instantiation, how could they possibly exert causal influence on bodies and bodies communicate sensation on minds? In other

words, how can two substances with such diverse natures causally interact? How could it tell your legs to move when you want to walk, and how can any physical stimulation reflect in your mind and become sensation (Elisabeth 96)? Given 15th century physics, this critique appeared to be highly damning, with Descartes and his contemporaries offering many subpar responses. Descartes asserts that the “pineal gland” does so magically. George Berkeley pivots metaphysically and denies the existence of the physical world: all things are ideas. Gottfried Leibniz goes even further with his idea of pre-established harmony, which claims that God synchronizes all thoughts and actions. The former is unattractive since it doesn’t actually respond to the how of the question and latter two are unattractive because they come with heavy theological baggage or deny free will outright.

Contemporary responses, however, make this problem much less damning. For one, Princess Elisabeth’s “mechanistic” mode of causation, which requires one object to push or pull upon another, is no longer believed to be the only mode of causation. Gravity, for instance, does not require one object to physically make contact with another object but is able to nonetheless exert influence. Thus, why can’t soul body interaction simply fall into another sort of non-mechanistic causation? Physicist Henry P. Stapp, for instance, argues that quantum mechanics already requires non-mechanistic elements since physical laws do not determine which measurement is performed or when state reduction occurs. For Stapp, conscious intention supplies this selection by choosing which quantum question is posed to the brain’s physical state, after which nature determines the outcome probabilistically. Thus, soul-body interaction does not occur in any classical mechanistic sense but operates through law governed fundamental quantum relations. Dualists are not limited to just Stapp’s quantum theory, however, as there are many other accounts of non mechanistic causation.

A more modern variation of the interaction problems centers on the fundamental law of conservation of energy. For a soul to give an order, it would need to influence the particles

of the brain. However, since brain particles are physical, and to alter the trajectory of any physical entity requires acceleration and thus energy, the soul would need to expend some amount of energy, but whence does this energy come? The soul by definition has no mass, no charge, no physical energy. Similarly, when souls are acted upon to receive sensory feedback, the brain would similarly need to expend energy to convey this sensation to the soul (Dennett, *Consciousness Explained*). However, the soul is immaterial so how could it receive this energy? In either world, critics contend that total energy is lost from this world.

While the Conservation Law version of the interaction problem might seem at first devastating, it rests upon a flawed interpretation of the Law of Conservation of Energy. Although the law states that total energy must stay the same, it does not dictate what kind of energy exists. As such, it is not inconceivable that “mental” or “soul” energy exists and that mind-body interaction involves the conversion of one kind of energy into another. William D. Hart does just that and argues for psychophysical causation to bridge the gap between mind and body. C.D. Broad gives an alternative account that suggests that a soul could act by simply redistributing the brain’s energy without changing its quantity. While Broad gives no account as to how this may occur, quantum theories like that given by Stapp could be the precise explanation. In summary, the Conservation of Energy critique falls short of satisfaction because it asserts without evidence that mental causation must require additional energy in the system and that only some types of energy exist.

Dualism matters in debates about consciousness because it gives us principled grounds for rejecting certain extreme hypotheticals, such as the one Daniel Dennett presents in “Where Am I?” (Dennett, “Where Am I?”). Dennett, working from functionalist assumptions closely aligned with the Computational Theory of Mind, suggests that a computer program which perfectly replicates the causal and functional organization of a human brain could itself be conscious. On this view, a person could alternate between control

by their biological brain and a computer duplicate without being able to detect any difference in experience. Dualism resists this conclusion by denying that consciousness is exhausted by functional organization alone, and thus rejects the claim that perfect functional similarity suffices for genuine subjective continuity. Moreover, Dennett's thought experiments also raises serious questions about who and what "I" is if one's mind (brain) can be duplicated. Dualism shuts this down as the soul is where the "I" resides and to our best knowledge, souls are unique.

A more contemporary theory of consciousness is Integrated Information Theory (IIT), developed by neuroscientist Giulio Tononi. IIT posits that consciousness arises from a system possessing specific, maximally integrated information about itself, including potential past and future states. IIT highlights the necessity of self-information, integration (where the system's total information exceeds the sum of its parts), and maximality (the system must have a higher level of integrated information, denoted as Φ , than any individual part). For instance, a subway app providing real-time locations integrates more relevant information than a novel, which, despite its detailed narrative, offers limited self-information.

Unlike other physical theories that attempt to arrive at the existence of consciousness from physical truths, IIT assumes the existence of consciousness, relies on intuition to posit various mental axioms, and derives the necessary qualities a physical system would need in order to account for it. For its first mental axiom, IIT claims that phenomenal experiences exist for itself and only itself, and its third axiom stipulates that conscious experience specifies a unique scene that's different from other scenes. From the fact that experiences exist only for themselves, IIT claims that the physical systems that produce consciousness must have information about itself, and from the fact that consciousness has a unique scene, IIT also claims that the corresponding physical system must rule out possible states from its total collection of states. This is the groundwork for IIT's particular definition of information.

Similarly, IIT also derives the importance of integration and maximality from its mental axioms (Tononi).

Now some might dismiss Cartesian dualism as a form of metaphysical make-believe, citing our lack of empirical evidence for an immaterial soul. Professor Greene certainly appears to belong to this camp. However, this line of criticism loses its force once we notice that many physicalist theories of consciousness aimed at solving the hard problem rely on the same kind of epistemic move they accuse dualists of making.

Consider Integrated Information Theory (IIT). IIT focuses on identifying and quantifying the integration of information within systems, proposing that systems with high integration possess higher levels of consciousness. However, this approach essentially correlates certain physical states with consciousness without explaining the underlying reason why these states should produce qualia. IIT's reliance on axiomatic assertions about what consciousness must be like does not bridge the explanatory gap between the physical and the experiential. Consciousness is not derived from physical principles but presupposes it assigning it to systems that satisfy its axioms without offering empirical evidence for why those axioms should generate qualia.

Similarly, while Professor Greene does not put forth a complete account of his favorite theory of consciousness, he seems to fall into the camp of what is called property dualism. Instead of assuming that there are two substances, one conscious (mind) one not (body), property dualists believe that there is only one substance (body) that just so happens to have the property of consciousness in addition to physical ones. This might seem more scientifically respectable, but commits the same epistemic transgressions as the Cartesian Dualism it critiques. I fail to see how the assertion that physical systems have mental properties, without explaining how or why those properties arise, is meaningfully better than

the Cartesian Dualist assertion of a new substance. In both theories, a primitive feature of our world is modified without independent empirical justification.

Given the uncertainty of the philosophy of consciousness, there may be no theoretic framework that allows us to predict whether an AI system is conscious or not. We may never arrive at the point where we will be able to definitively know whether an AI is conscious or not simply by scrutinizing its underlying structure or chemical makeup. This, however, does not mean that we should give up and blindly decree that they will forever remain unconscious. Afterall, we are never able to indisputably verify any other human's consciousness but we still treat them as such. Instead, as suggested by futurist Susan Schneider, there are behavioral questions that may be administered to elucidate the mental state of machines. While this is not foolproof, given the possibility of mechanical zombies, it may be a good starting point. For instance, humans are able to understand "body switch stories" pretty well (Schneider 51). In both *Freaky Friday* and *Your Name*, characters switch bodies. This is due to our shared understanding that it is possible to separate our mind from our physical body. However, it may be difficult for an inwardly dark computer to grasp the same idea. Therefore, we could ask questions like "Could you or your inner processes be in a separate location from the computer? From any computer? Why or why not?", "Would you still survive if we deleted every copy of your program?", or see how it grasps general concepts like reincarnation (Schneider 52,55). Finally, we could also implement a variation of the famous "Mary in a Red Room" experiment on the AI. We can do this if we isolate the AI in a certain environment, give it all the physical information surrounding a certain thing that it has not experienced or sensed, and then expose this new thing directly to it. If the machine can sense color but has not sensed red ever, for instance, we may describe to it all the physical qualities of the color red, and then allow it to finally sense red (Jackson 129-131). If it reacts like it is surprised or it has come into contact with something new, we may conclude

that it may have qualia, while if it is not surprised or acts indifferently, it probably does not. All of these questions, however, operate under the assumption that whatever AI is being tested, it must have been trained or “raised” under tight informational secrecy. What this means is that they must not be exposed to any information or language surrounding consciousness, the soul or other things like it, or else we run into the danger of creating mechanical zombies like ChatGPT and other LLM’s, thereby rendering the consciousness test inert.

The ultimate goal of this essay is not to once and for all confer upon its reader the newly discovered answer of how consciousness arises or to definitively predict whether machines are or will be conscious. Instead, I hope to demonstrate just how complex consciousness is and how most people do not grasp the full gravity and complexity of it. In fact, it is not just “most people” who lack an understanding of it but even philosophers themselves. Although philosophy of mind and questions of consciousness are perennial questions that continue to fascinate us today, they seem to have yielded more questions than answers. This is inadequate because as moral philosopher Joshua Shepherd points out, phenomenal consciousness has “non derivative” value in our moral system (Shepherd 11). Therefore, we risk enslaving or committing serious moral transgressions on AI if we fail to identify when and if they become conscious. The stakes of unraveling the mysteries of phenomenal consciousness have never been higher.

Works Cited

- Avramides, Anita. "Other Minds." *Stanford Encyclopedia of Philosophy*, 2 May 2019, plato.stanford.edu/entries/other-minds/. Accessed 13 Apr. 2024.
- Blaer, Paul. "Introduction to Computer Science and Programming in Java." Columbia University, 6 Sept. 2023. Speech.
- Broad, C. D. *The Mind and Its Place in Nature*. Routledge & Kegan Paul, 1925.
- Byrne, Alex. "Inverted Qualia." *Stanford Encyclopedia of Philosophy*, 10 Nov. 2004, plato.stanford.edu/entries/qualia-inverted/. Accessed 13 Apr. 2024.
- Chalmers, David J. "FACING UP TO THE PROBLEM OF CONSCIOUSNESS." *Journal of Consciousness Studies*, vol. 2, no. 3, 1995, pp. 200-19.
- Descartes, René. "Meditation Six: Concerning the Existence of Material Things, and the Real Distinction between Mind and Body." 1641. *Modern Philosophy: An Anthology of Primary Sources*, edited by Roger Ariew and Eric Watkins, 3rd ed., Hackett Publishing Company, 2019, pp. 61-68.
- Dennett, Daniel C. *Consciousness Explained*. Little, Brown and Co., 1991.
- . "Where Am I?" *The Library of Consciousness*, 13 July 1981, Organism.Earth, <https://www.organism.earth/library/document/where-am-i>. Accessed 21 Dec. 2025.
- Elisabeth of Bohemia. "Elisabeth and Descartes, Correspondence." 1643. *Modern Philosophy: An Anthology of Primary Sources*, edited by Roger Ariew and Eric Watkins, 3rd ed., Hackett Publishing Company, 2019, pp. 93-99.
- Gennaro, Rocco, editor. *Internet Encyclopedia of Philosophy*.
iep.utm.edu/computational-theory-of-mind/#:~:text=The%20Computational%20Theory%20of%20Mind,working%20hypothesis%20of%20cognitive%20science. Accessed 13 Apr. 2024.

- Guinness, Henry. "How does ChatGPT work?" *Zapier*, 6 Sept. 2023, zapier.com/blog/how-does-chatgpt-work/. Accessed 13 Apr. 2024.
- Hart, W. D. *The Engines of the Soul*. Cambridge University Press, 1988.
- Jackson, Frank. "Epiphenomenal Qualia." *The Philosophical Quarterly* (1950-), vol. 32, no. 127, 1982, pp. 127-36. *JSTOR*, <https://doi.org/10.2307/2960077>. Accessed 13 Apr. 2024.
- Kirk, Robert. "Zombies." *Stanford Encyclopedia of Philosophy*, 8 Sept. 2003, plato.stanford.edu/entries/zombies/#IdeaZomb. Accessed 20 Dec. 2025.
- Levin, Janet. "Functionalism." *Stanford Encyclopedia of Philosophy*, edited by Edward N. Zalta and Uri Nodelman, 24 Aug. 2004, plato.stanford.edu/entries/functionalism/. Accessed 20 Dec. 2025.
- Robinson, Howard. "Dualism." *Stanford Encyclopedia of Philosophy*, 19 Aug. 2003, plato.stanford.edu/entries/dualism/. Accessed 20 Dec. 2025.
- Schneider, Susan. *Artificial You*. Princeton University Press, 2019.
- Scott, Cameron. "Study finds ChatGPT's latest bot behaves like humans, only better." *Stanford School of Humanities and Sciences*, 22 Feb. 2024, humsci.stanford.edu/feature/study-finds-chatgpts-latest-bot-behaves-humans-only-better#:~:text=In%20the%20study%2C%20ChatGPT's%20version,and%20motherboards%2C%20above%20version%203. Accessed 20 Dec. 2025.
- Searle, John R. "Chinese Room Argument." *The MIT Encyclopedia of the Cognitive Sciences*, edited by Robert A. Wilson and Frank C. Keil, Cambridge Massachusetts, MIT Press, 1999, pp. 115-16.
- Shepherd, Joshua. *Consciousness and Moral Status*.
- Stapp, Henry P. "Quantum Interactive Dualism: An Alternative to Materialism." *Journal of Consciousness Studies*, vol. 12, no. 11, 2005, pp. 43–58.

Tononi, Giulio. "An information integration theory of consciousness." *BMC Neuroscience*, vol. 5, no. 1, 2 Nov. 2004, <https://doi.org/10.1186/1471-2202-5-42>. Accessed 20 Dec. 2025.

Turing, Alan. "Computing Machinery and Intelligence." *Mind*, vol. 59, 1950, pp. 433-60.